Adaptive Stochastic Matching: An Expected Query Bound and the Cost of Mandatory Batching

Maximilian Kornstein

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Abstract

We study adaptive edge discovery in a known finite bipartite graph whose edges are independently present with probability 1/2. Queries are charged at both endpoints, successful edges may be retained without commitment, and the objective is expected matching size relative to the expected full-information optimum. We give two results. First, the established optimistic maximum-matching-round policy, run for R rounds, has expected relative loss at most $2\log 3/(R\log(3/2))$. It therefore obtains a $(1-\varepsilon)$ approximation with a hard per-vertex query cap $\lceil 6/\varepsilon \rceil$ on every finite input. The proof tracks evolving vertex covers indexed by subsets of one minimum cover of the host graph. Second, on an explicit family with accuracy parameter ε_N, every policy that must finish each proposed optimistic matching batch needs $\Omega(1/(\varepsilon_N \log^2(1/\varepsilon_N)))$ queries at some vertex, whereas interior-first screening achieves the same expected accuracy with $O(\log(1/\varepsilon_N))$ queries per vertex. The screening policy also uses optimistic maximum-matching proposals, but cancels a proposal after a negative answer. Thus within-proposal stopping can change endpoint query cost exponentially on this family. The results concern mathematical discovery complexity, not empirical relationship prediction.

1 Introduction

When testing an edge is costly, a large matching in the underlying graph is only useful if its edges can be discovered. Stochastic matching formalizes this distinction: a known host graph specifies the possible edges, and queries reveal independent edge realizations. A local budget limits the number of tests involving each vertex. We consider deferred matching, in which positive answers need not be accepted immediately and the final matching is chosen from all discovered positive edges.

This paper separates two features of adaptive discovery. Repeatedly proposing a maximum matching of all edges not yet ruled out gives a simple global approximation guarantee. Requiring every proposed batch to be completed, however, can consume a scarce endpoint budget on proposals that an earlier answer has already made unhelpful. We analyze both effects under the same independent-edge model and the same hard local budget.

Our first result is a direct finite-input expected analysis of the familiar optimistic matching-round algorithm. With $\mu = \mathbb{E}[\nu(G_{1/2})]$, its output A_R satisfies

$$\mathbb{E}[\nu(G_{1/2}) - A_R] \le \frac{2\log 3}{R\log(3/2)}\mu < \frac{5.42}{R}\mu. \tag{1}$$

Every vertex participates in at most R queries on every execution. There is no minimum-size condition on the host matching. The algorithm does not enumerate covers: the cover family appears only in the analysis.

Our second result identifies a limitation of mandatory batches. On a family of parallel chains of dense bipartite blocks, every mandatory-batch policy attaining the specified accuracy needs a hub budget linear in the number N of chains. Screening interior edges before committing hub queries uses a budget logarithmic in N. Both bounds hold on the initially fresh product distribution. The lower bound allows arbitrary adaptive maximum-matching tie-breaking, so changing the tie-breaking rule cannot remove this separation.

These are two compatible results: the global upper bound is inverse in accuracy, and the construction requires nearly inverse-accuracy budget from mandatory batches. The construction does not impose that requirement on all adaptive policies. In particular, this paper does not establish a logarithmic-budget algorithm on arbitrary bipartite hosts.

1.1 Relation to prior work

Assadi, Khanna, and Li [1] study adaptive stochastic matching through optimistic maximum-matching rounds, matching covers, and vertex sparsification. Their stated adaptive bound is $O(\log(1/(\varepsilon p))/(\varepsilon p))$ queries per vertex, with a high-probability guarantee. Maehara and Yamaguchi [2] develop a broader dual-witness framework for stochastic packing. Their Corollary 5.1 includes finite bipartite matching and gives a $(1-\eta)$ approximation with probability at least $1-\eta$ using $O(\log(1/(\eta p))/(\eta p))$ queries per vertex. Integrality permits one optimistic matching per round in the bipartite unweighted case.

For comparison with our expected objective, let $m = \nu(G)$ and $\mu = \mathbb{E}[\nu(G_p)] \geq pm$. A guarantee with failure probability η and approximation $1-\eta$ implies

$$\mathbb{E}[A] \geq (1-\eta)\mu - \eta m \geq \big(1-\eta(1+1/p)\big)\mu.$$

Choosing $\eta = \varepsilon p/(1+p)$ gives an $O(\log(1/\varepsilon)/\varepsilon)$ bound at fixed p from that cited result. Equation (1) removes this logarithmic factor in the expected guarantee at $p = 1/2$. We attribute the algorithm to the existing literature; the contribution presented here is the analysis and explicit bound. We do not assert historical priority over every variant of stochastic matching.

The block-chain construction in our separation is adapted from the resistor construction described by Glebov, Luria, and Simkin [3, Appendix A], who attribute it to Goel, Kapralov, and Khanna [4]. We separate the number of chains from the block width and analyze adaptive query policies. We do not claim the underlying gadget as a new construction.

2 Model and notation

Let $G = (L \cup R, E)$ be a known finite simple bipartite graph. Every edge $e \in E$ is independently present with probability $p \in (0,1)$, giving the random realized graph G_p. Write $\nu(F)$ for the cardinality of a maximum matching in a graph F, and put

$$m = \nu(G), \qquad \mathrm{OPT} = \nu(G_p), \qquad \mu = \mathbb{E}[\mathrm{OPT}].$$

The principal results fix $p = 1/2$. A fixed host maximum matching gives $\mu \geq pm$.

An adaptive policy queries an unqueried edge and observes whether it is present. Its decisions can depend on all previous answers and independent internal randomness. A query costs one unit at each endpoint. A hard per-vertex budget B means that every vertex is queried at most B times on every execution, not merely on average. There are no repeated queries. Let H contain all observed positive edges, and let U contain all host edges except observed negatives. Thus $H \subseteq G_p \subseteq U$. The

policy outputs a matching in H; it may retain overlapping positive edges until the final decision. Its quality is measured by

$$\mathbb{E}[\nu(H)] \geq (1-\varepsilon)\mu, \tag{2}$$

where outputting a maximum matching in H is always permitted. This is a ratio of expectations, not an expected pointwise ratio.

A *mandatory optimistic batch* chooses a maximum matching of U and queries every previously unqueried edge in it. The selected batch cannot be canceled in response to one of its answers. Batch choices may be adaptive and randomized; no assumption is made about tie-breaking. A policy may stop between batches. In contrast, a *screening policy* may cancel a proposal immediately after a negative answer and recompute.

3 An expected inverse-accuracy query bound

The algorithm starts with $U_0 = G$ and H_0 empty. In each of R rounds, it chooses any maximum matching M_i of U_{i-1}, queries all unknown edges of M_i, adds positive answers to H, and deletes negative answers from U. Let H_R, U_R be the final graphs, and let $A_R = \nu(H_R)$.

Theorem 3.1. *For every finite bipartite host G with independent edge probability $p = 1/2$, and every integer $R \geq 1$, this policy has hard per-vertex query cap R and satisfies*

$$\mathbb{E}[\mathrm{OPT} - A_R] \leq \frac{2\log 3}{\log(3/2)}\frac{\mu}{R}.$$

Consequently $R = \lceil 6/\varepsilon \rceil$ gives (2) for every $0 < \varepsilon < 1$. The policy runs in polynomial time in the graph size and R.

Each round queries a matching, so its contribution to every endpoint's query count is at most one. The computational claim follows from polynomial-time bipartite maximum matching. It remains to prove the expected loss bound. We give the analysis for general p until its final specialization. The case $m = 0$ is immediate; assume $m > 0$.

3.1 A canonical family of evolving covers

Fix a minimum vertex cover C of the host G. By Kőnig's theorem, $|C| = m$, and $V(G) \setminus C$ is an independent set. For $S \subseteq C$ and an observed-positive graph H, define

$$F_S(H) = S \cup \big(N_H(C \setminus S) \setminus C\big). \tag{3}$$

For fixed S this set grows monotonically as positive edges are revealed.

Lemma 3.2. *The set $F_S(H)$ is a vertex cover of H if and only if $H[C \setminus S]$ has no edge. Every minimum vertex cover W of H satisfies $W = F_{W\cap C}(H)$.*

Proof. Every edge of H touches C. An edge touching S is covered; an edge from $C \setminus S$ to outside C is covered at its outside endpoint by (3). An edge within $C \setminus S$ is uncovered, and these exhaust the possibilities.

For a vertex cover W, put $S = W \cap C$. There is no H-edge within $C \setminus S$, since neither endpoint would be in W. Every neighbor outside C of $C \setminus S$ must belong to W. Hence $F_S(H)$ is a vertex cover contained in W. If W is minimum, that containment must be equality. □

3.2 Stopped failure counts

For each fixed $S \subseteq C$, introduce an analysis track. It is active initially unless $|S| = m$. Stop it permanently as soon as $F_S(H)$ ceases to be a vertex cover or reaches size m. While active, count a query precisely when both queried endpoints lie outside the current $F_S(H)$. Let X_S be the number of negative answers among its counted queries before stopping or termination of the algorithm.

Lemma 3.3. *For $1 < z < 1/(1-p)$, set $b = p/[1-(1-p)z]$. Then*

$$\mathbb{E}[z^{X_S}] \le b^{m-|S|}. \tag{4}$$

Proof. Whether a fresh query is counted is determined before its answer is observed. Conditional on the entire past, its answer is an independent Bernoulli(p) outcome. A counted positive either joins two vertices in $C \setminus S$, invalidating the cover, or joins a vertex there to a previously unlisted vertex outside C, increasing the cover size by one. No third case is possible because C covers the host. Thus the track stops by its $(m-|S|)$th counted positive. Other observations cannot delay this stopping rule.

The counted outcome sequence, extended after stopping by independent coins, is an i.i.d. Bernoulli(p) sequence. Therefore X_S is bounded by the number of failures before $m-|S|$ successes. The probability generating function for failures before one success is $p/[1-(1-p)z]$; multiplication over the successes gives (4). For $S = C$, the track is inactive and $X_S = 0$. This argument is marginal for each track and requires no independence between different tracks. □

3.3 The final gap forces repeated failures

Write $r = \nu(H_R)$, $\tau = \nu(U_R)$, and $D = \tau - r$. Since $H_R \subseteq G_p \subseteq U_R$, pathwise

$$0 \le \mathrm{OPT} - r \le D. \tag{5}$$

Lemma 3.4. *On every execution,*

$$RD \le \max_{S \subseteq C} X_S. \tag{6}$$

Proof. If $D = 0$ the assertion is immediate. Otherwise $r < \tau \le m$. Choose a minimum vertex cover W of H_R, of size r, and set $S_* = W \cap C$. By theorem 3.2, $F_{S_*}(H_R) = W$. At every earlier time this evolving cover is a subset of W, remains valid, and has size below m. Its track is therefore always active.

The optimistic graphs only lose edges. In every round, $|M_i| \ge \tau$. At most $|W| = r$ edges of a matching touch W, so M_i has at least D edges with both endpoints outside the final W. Each such edge must be fresh in that round: a known negative would already have been deleted, and a known positive would lie in H_R and be covered by W. It must receive a negative answer, again because W covers H_R. Since the current $F_{S_*}(H)$ is a subset of W, this failure is counted by X_{S_*}. Failed edges are deleted and cannot be counted again in a later round. Summing over the rounds gives $RD \le X_{S_*}$, which proves the claim. □

Completion of the proof of theorem 3.1. Let $X = \max_{S \subseteq C} X_S$. From theorem 3.3,

$$\mathbb{E}[z^X] \le \sum_{S \subseteq C} \mathbb{E}[z^{X_S}] \le \sum_{S \subseteq C} b^{m-|S|} = (1+b)^m.$$

Jensen's inequality gives

$$\mathbb{E}[X] \le \frac{\log \mathbb{E}[z^X]}{\log z} \le \frac{m \log(1+b)}{\log z}.$$

Combining (5) and (6),

$$\mathbb{E}[\mathrm{OPT} - A_R] \le \frac{m \log(1+b)}{R \log z}. \tag{7}$$

For $p = 1/2$ and $z = 3/2$, we have $b = 2$ and $m \le 2\mu$. Substitution proves the theorem; $2 \log 3 / \log(3/2) < 5.42 < 6$ gives the stated round count. □

Proposition 3.5 (General edge probability)**.** *For any fixed $p \in (0,1)$, the same policy and every $1 < z < 1/(1-p)$ satisfy*

$$\mathbb{E}[\mathrm{OPT} - A_R] \le \frac{\log(1 + p/[1-(1-p)z])}{pR \log z} \mu.$$

Proof. Use $m \le \mu/p$ in (7). □

Taking $z = 1/(1-p/2)$ gives $b = 2-p$. This specialization has $O(1/(p^2 R))$ relative loss as $p \downarrow 0$; the main improvement asserted here is in accuracy dependence at fixed $p = 1/2$.

4 The cost of mandatory batches

The next result isolates the effect of finishing a proposal after one of its edges has failed. A *mandatory batch policy* maintains the optimistic graph of all edges not yet found negative. Each round selects a maximum matching in that graph and queries every fresh edge in the selected matching. It may stop between rounds, and its choice of matching may depend arbitrarily on the past and on private randomness. It cannot cancel a selected query in response to another answer within the same batch.

Theorem 4.1 (Separation from interior-first screening)**.** *Let $r \ge 2$ be an integer and set*

$$N = 4^r, \qquad \ell = \log_2 N, \qquad d = 16 \log_2 N, \qquad n = Nd\ell + 1,$$

$$c_0 = 1 - e^{-1/2}, \qquad \epsilon_N = \frac{c_0}{4n}.$$

There is a finite simple bipartite host G_N, with n vertices on each side, whose independently retained edges have probability $1/2$, such that:

1. *Every mandatory batch policy satisfying*

   $$\mathbb{E}[\nu(H)] \ge (1 - \epsilon_N)\mathbb{E}[\nu((G_N)_{1/2})]$$

   and a hard query cap B at every vertex has $B \ge c_0 N$.

2. *A polynomial-time policy using optimistic maximum-matching proposals, but cancelling a proposal on its first negative answer, attains the same expected guarantee with a hard cap d at every vertex.*

Consequently, along this family the necessary budget for mandatory batches is

$$\Omega\left(\frac{1}{\epsilon_N \log^2(1/\epsilon_N)}\right),$$

whereas the screening policy uses $O(\log(1/\epsilon_N))$ queries per vertex.

The construction uses the resistor-chain structure described in Appendix A of Glebov, Luria, and Simkin [3], where it is attributed to Goel, Kapralov, and Khanna. Here the block width and number of chains are independent parameters. The host need not be regular. We use this established construction to compare discovery policies.

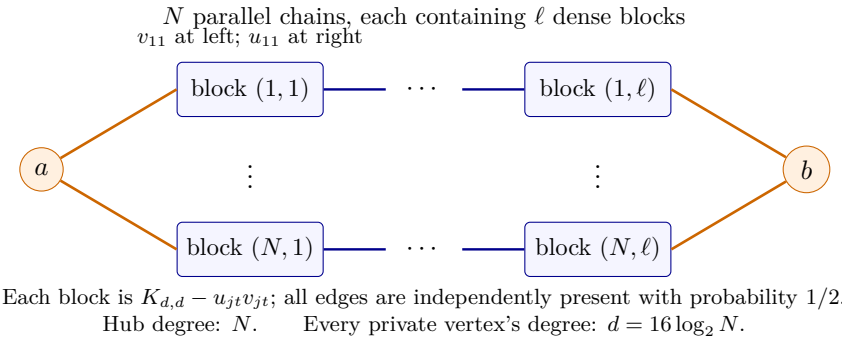


Figure 1: The separation host, shown schematically. Lines between blocks are single connector edges; a block contains d vertices on each side. A perfect matching must use the entire connector chain for exactly one of the N chains. Screening queries nonhub edges before the orange hub edges and cancels a proposal at its first failure. The figure is structural, not a plot of experimental results.

4.1 The host and its perfect matchings

Put a hub a on the left and a hub b on the right. For every $(j,t) \in \{1,\dots,N\} \times \{1,\dots,\ell\}$, create a private block with d left vertices U_{jt} and d right vertices V_{jt}. Distinguish ports $u_{jt} \in U_{jt}$ and $v_{jt} \in V_{jt}$. The internal block is $K_{d,d}$ with the edge $u_{jt}v_{jt}$ removed. Add the connectors

$$av_{j1}, \qquad u_{jt}v_{j,t+1} \quad (1 \le t < \ell), \qquad u_{j\ell}b.$$

Every private vertex has degree exactly d: a port has $d-1$ internal edges and one connector, while an ordinary vertex has d internal edges. Each hub has degree N. All edges, including block edges and connectors, are initially unknown and independently positive with probability $1/2$.

The construction is illustrated in Figure 1.

In a perfect matching, each block's entering and leaving connector indicators must agree. Otherwise, deleting vertices matched externally would leave unequal numbers of its left and right vertices to match internally. Since the two hubs are matched once, every perfect matching uses all $\ell+1$ connectors of exactly one chain and no connectors of the other chains. Conversely, the deterministic host has a perfect matching: activate one chain, match its blocks after deleting their ports, and match every other block internally. Thus, writing $\mu_N = \mathbb{E}[\nu((G_N)_{1/2})]$, the surviving edges of a fixed host perfect matching give

$$\frac{n}{2} \le \mu_N \le n. \tag{8}$$

4.2 The realized graph has a perfect matching with constant probability

Lemma 4.2 (An elementary block bound)**.** *Let $k \ge 32$. Retain the edges of $K_{k,k}$ independently with probability $1/2$, optionally with one edge deleted deterministically in advance. The probability that the resulting graph has no perfect matching is at most $8k^2 2^{-k/2}$.*

Proof. Hall's theorem gives a left set of size s whose neighborhood lies in some right set of size $s-1$. The $s(k-s+1)$ edges between that left set and the complement of the right set must all be absent. A single deterministic deletion increases the probability by at most a factor of two. A union bound therefore gives

$$2\sum_{s=1}^{k} \binom{k}{s}\binom{k}{s-1} 2^{-s(k-s+1)}.$$

Put $x = k^2 2^{-k/2}$. For $s \le \lceil k/2 \rceil$, the summand without its leading factor of two is at most x^s. For larger s, substitute $t = k - s + 1$ and use the same bound, since the summand is invariant under that substitution. The whole expression is at most $4 \sum_{s \ge 1} x^s = 4x/(1-x)$. For $k \ge 32$, $x \le 1/2$, giving the claimed bound. □

Let $\mathcal{E}$ be the event that every block has both an internal perfect matching and a perfect matching after deleting its two ports. The latter graph is a fair random $K_{d-1,d-1}$. Since $d \ge 64$, Lemma 4.2 and a union bound give

$$\begin{aligned} \mathbb{P}(\mathcal{E}^c) &\le \eta_N := 8N\ell \left[d^2 2^{-d/2} + (d-1)^2 2^{-(d-1)/2} \right] \\ &\le \frac{6144(\log_2 N)^3}{N^7} < \frac{1}{4}. \end{aligned} \tag{9}$$

For the last inequality, the displayed upper bound is decreasing for $\log_2 N \ge 4$ and is already less than $1/4$ at $N = 16$.

The event $\mathcal{E}$ involves only block internals. Independently, a given chain has all its connectors positive with probability

$$q = 2^{-(\ell+1)} = \frac{1}{2N}. \tag{10}$$

These events are independent across chains. On $\mathcal{E}$, any such chain extends to a global perfect matching, so

$$\mathbb{P}\big(\nu((G_N)_{1/2}) = n\big) \ge (1 - \eta_N) \left[1 - \left(1 - \frac{1}{2N} \right)^N \right] \ge \frac{3c_0}{4}. \tag{11}$$

4.3 The mandatory-batch lower bound

While the optimistic matching number is n, every proposed matching is perfect and therefore selects the complete connector set of exactly one chain. At the first proposal selecting that chain, all its connectors are fresh: every earlier perfect proposal either used the entire chain or used none of it. The complete connector vector is independent of the preceding transcript, even if that transcript contains internal information about the chain. Its conditional probability of being entirely positive is q.

Each newly selected chain spends one fresh query at a. Thus a policy with hard cap B can select at most B new chains while the optimistic rank is n. A union bound for this adaptively selected sequence yields

$$\mathbb{P}\big(\nu(H) = n\big) \le Bq = \frac{B}{2N}. \tag{12}$$

Indeed, a discovered perfect matching needs an entirely positive connector vector from one of those chains. Reusing a chain does not add a new candidate. If the optimistic rank ever drops below n, the realized graph itself has no perfect matching, so later proposals cannot add another route to the event in (12). The bound holds for every history-dependent tie-breaking rule and any number of completed batches.

Since $H \subseteq (G_N)_{1/2}$ and matching numbers are integers,

$$\mathbb{E}[\nu((G_N)_{1/2}) - \nu(H)] \ge \mathbb{P}\big(\nu((G_N)_{1/2}) = n\big) - \mathbb{P}\big(\nu(H) = n\big).$$

The requested approximation allows absolute loss at most $\epsilon_N \mu_N \le c_0/4$, by (8). Equations (11) and (12) therefore imply

$$\frac{3c_0}{4} - \frac{B}{2N} \le \frac{c_0}{4}, \qquad \text{hence} \qquad B \ge c_0 N.$$

4.4 Screening the same type of proposal

Consider the following policy, with $T = 16$.

1. Choose any maximum matching Q in the current optimistic graph. Query its fresh edges not incident to a or b first. On a negative answer, cancel the rest of the proposal, delete the failed edge, and return to this step.

2. Once every nonhub edge of Q is known positive, query its remaining fresh hub edges. There are at most two. Call this a *hub trial*. On the first negative answer, cancel the trial and recompute a proposal.

3. A fully verified proposal is returned immediately. After T failed hub trials, return a maximum matching of all discovered positive edges.

A proposal with no fresh edges is already fully verified. A proposal with no fresh hub edges is fully verified as soon as its interior queries succeed, and uses no hub trial.

Successful queries do not alter the optimistic graph. Thus a fully verified proposal is still maximum in a graph containing the entire realization, and certifies an exact answer. Every nonterminal action queries a new edge, so the algorithm terminates after finitely many queries and polynomially many ordinary matching computations.

When the first hub trial begins, let τ be the size of its proposal. All but at most two of its edges are known positive. The retained graph therefore already contains a matching of size at least $\tau - 2$, while the realized optimum is at most τ. Positives are retained forever, so the eventual loss is at most two on every execution that reaches a hub trial. Executions terminating before a hub trial are exact.

Conditioned on the full history at the start of any hub trial, its one or two fresh edges have independent fair outcomes. With conditional probability at least $1/4$, they all succeed and certify exactness. Intervening interior screening can also certify exactness. Iterating these conditional bounds shows that the probability of reaching T failed trials is at most $(3/4)^T$. Consequently,

$$\mathbb{E}[\nu((G_N)_{1/2}) - \nu(H)] \le 2(3/4)^{16} < \frac{1}{24} < \frac{c_0}{8} \le \epsilon_N \mu_N. \tag{13}$$

Here $c_0 > 1/3$ follows from $e^{1/2} > 3/2$, and the last inequality follows from (8).

Each trial makes at most one query at each hub, so their query counts are at most 16. Every private vertex has total host degree d and no edge is queried twice. Its total count, including any hub connector, is therefore at most d. As $d \ge 64$, the hard cap for the entire policy is d. This proves the second assertion of Theorem 4.1. Finally, $n = 16N(\log_2 N)^2 + 1$ implies the asserted dependence on ϵ_N.

Remark 4.3 (Scope of the separation). Both policies face the same fully fresh distribution, expected realized optimum, accuracy requirement, and query charges at both endpoints. The screening policy still proposes optimistic maximum matchings and works with any tie-breaking rule. Its distinguishing operation is cancelling selected hub queries after an interior failure. The theorem does not give a lower bound against arbitrary adaptive policies or a logarithmic upper bound on arbitrary hosts. The screening argument exploits the two-hub structure of this particular family.

5 Consequences and limits

The first result bounds an established algorithm without any dependence on the number of vertices in the query cap. Its proof uses only one host minimum cover and predictable fresh-edge outcomes.

Exponentially many analysis tracks are combined through a single exponential-moment sum; they are not states maintained by the algorithm.

The separation explains why the order of discovery can matter even when every proposal is a maximum matching. A mandatory batch commits scarce hub queries before knowing whether its interior connectors can succeed. Screening rejects unsuccessful interiors before spending that allowance. The same-family comparison concerns the same full-information oracle, the same accuracy, and budgets charged at both endpoints.

Neither theorem establishes a universally logarithmic query bound for adaptive policies. The separation is a lower bound for the mandatory class only. It also does not imply a uniform improvement in p over the cited work: the general-p analysis in theorem 3.5 has $O(p^{-2})$ dependence for small p. No assertion of globally optimal adaptive query complexity follows from these results.

Finally, the model contains neither preferences nor strategic choice. Its independent binary edges are assumed, rather than learned from people. It makes no claim about predicting compatibility, producing stable marriages, or improving real dating outcomes. Such conclusions would require a different model and empirical evidence. All results here are analytical; there are no human-subject data or experimental performance claims.

Availability and preparation. The accompanying source archive contains the complete LaTeX manuscript and its vector figure. No external data or simulations are required to reproduce the stated results. OpenAI Codex was used for mathematical derivation, drafting, and internal proof review. This preprint has not undergone peer review.

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